

Chapter 15: Data Through Pictures

Worksheet 15: Data Through Pictures

Learning Objective: Read, interpret and construct pictographs and bar graphs to represent and analyse simple data.

Worked Example:

A pictograph shows that each symbol represents 5 students. If a row has 4 symbols, how many students does it represent?

Solution: $4 \times 5 = 20$ students.

Questions (1–20)

1. In a pictograph, if one symbol = 10 books, how many books do 3 symbols represent?
2. What is the purpose of a “key” or “legend” in a pictograph?
3. In a bar graph, what does the height of a bar usually represent?
4. A bar graph shows fruit sales. The bar for “mango” reaches 25 on the y-axis. How many mangoes were sold?
5. Name one advantage of using a pictograph instead of just a list of numbers.
6. A pictograph of ice creams sold uses a key of 1 symbol = 4 ice creams. If Monday shows 3 symbols, how many ice creams were sold?

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7. A bar graph shows favourite sports of 40 students: Cricket-18, Football-10, Badminton-8, Others-4. Which sport is most popular?

 8. Using the data above, how many more students prefer cricket than badminton?

 9. A pictograph uses 1 symbol = 2 students for modes of transport to school. If “Bicycle” shows 6 symbols, how many students cycle to school?

 10. A bar graph shows rainfall (mm) for 4 months: Jan-20, Feb-15, Mar-35, Apr-25. Which month had the highest rainfall?

 11. Using the rainfall data above, find the total rainfall over the four months.

 12. A pictograph (1 symbol = 5 mangoes) shows mangoes sold over 5 days: Mon-4, Tue-3, Wed-6, Thu-2, Fri-5 symbols. Find the total mangoes sold over the week.

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13. From the data above, on which day were the fewest mangoes sold, and how many?
14. A class survey on favourite fruits: Apple-12, Banana-8, Mango-15, Orange-5. Describe how you would represent this in a bar graph, choosing a suitable scale.
15. If the total number of students above is 40, what fraction chose mango as their favourite?
16. A pictograph (1 symbol = 3 books) shows books read in a month: Asha-5, Ravi-3, Meena-4, Karan-2 symbols. Who read the most books, and how many?
17. A bar graph shows trees planted by 5 classes: A-22, B-18, C-30, D-15, E-25. Find the average number of trees planted per class.
18. Using the data above, if each tree costs Rs. 40 to plant, find the total cost for all 5 classes combined.

19. Conduct a small survey among 10 family members or friends about their favourite festival and represent your data using a simple pictograph or bar graph (or describe it in words/a table).

20. Find an example of data shown using pictures or bars in a newspaper, textbook, or app (like a weather app). Describe what it shows.

Reflection Question: Why do you think pictures and bars are used to show data instead of just writing numbers in a list? Write one reason in your own words.

Real-Life Application Question: Find one example at home or school where information is shown using a picture-based chart or graph (a notice board, newspaper, or food label). Describe what it shows.